

# Lower growth rates and earlier mortality of seedlings in a logged versus old-growth tropical forest

## Supporting Information

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### 1 Climber cutting

Three of the Sabah Biodiversity Experiment plots included in this study received a complete climber cutting treatment as part of a different experiment (plots numbered 5, 11 and 14), where all lianas  $\geq 10$  cm in height were cut at the base in 2011 and 2014. <sup>1</sup>

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<sup>1</sup>O'Brien, M. J. et al. Positive effects of liana cutting on seedlings are reduced during El Niño-induced drought. Journal of Applied Ecology 56, 891–901 (2019). doi: [10.1111/1365-2664.13335](https://doi.org/10.1111/1365-2664.13335)

Including the climber cutting treatment as a fixed effect in our growth and survival models did not change our results.

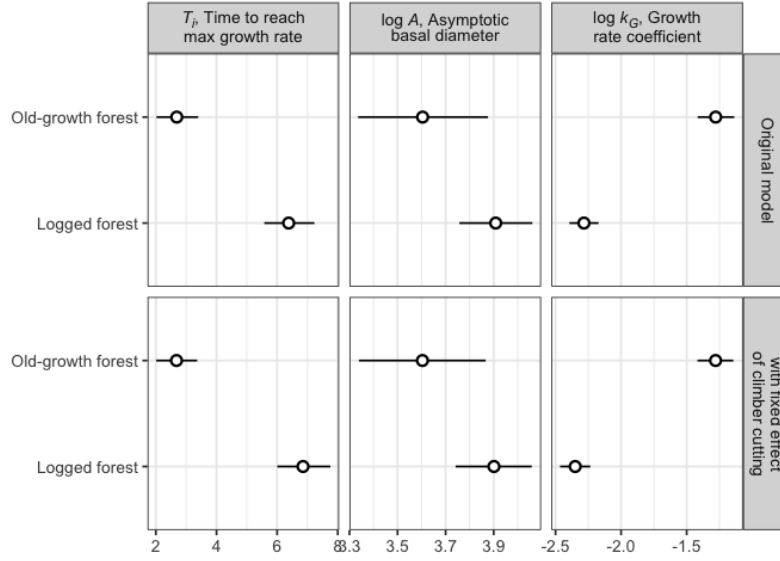


Figure 1: Comparison of parameter estimates for the growth model with and without a fixed effect of climber cutting. Facet columns indicate the parameters of the growth function with estimates for the old-growth and logged forest types on the  $y$  axis. Facet rows show results from the original model presented in the main text, and a model with the addition of a fixed effect for climber cutting treatment. Point shows the median of the posterior distribution with lines denoting the 95% credible interval.

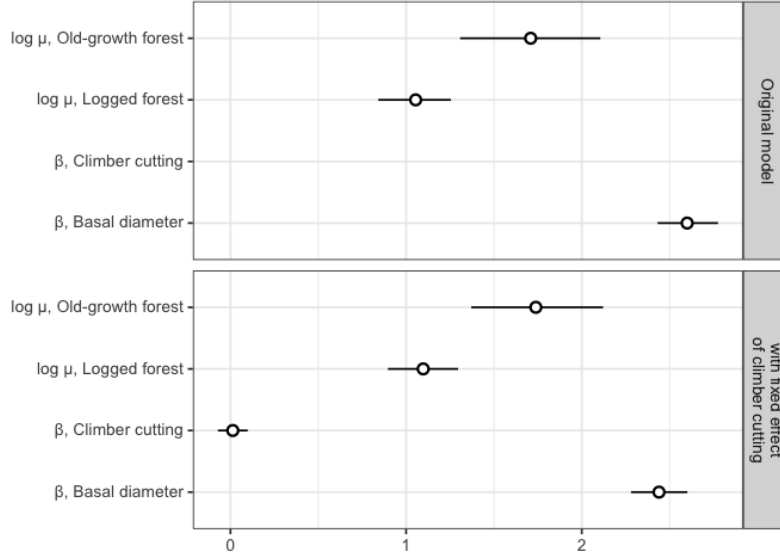


Figure 2: Comparison of parameters from survival models with and without a fixed effect of climber cutting. Facet rows show results from the original model presented in the main text, and a model with the addition of a fixed effect for climber cutting treatment. Point shows the median of the posterior distribution with lines denoting the 95% credible interval.

## 2 Species taxonomy

In the main text we use current accepted species names. Here, for reference, we list the old names, which are consistent with prior publications involving the Sabah Biodiversity Experiment. Species names were matched to a static copy of [The World Flora Online \(WFO\)](https://doi.org/10.1111/1365-3113.12511) v.2024.12 [zenodo.14538251](https://doi.org/10.1111/1365-3113.12511) using the function `WFO.match` from the R package `{WorldFlora}`. The function `WFO.one` was then used to find one unique matching name for each submitted name. using the argument `priority = "Accepted"`, it first limits candidates to accepted names, with a possible second step of eliminating accepted names that are synonyms.

Table 1: Taxonomy of species included in the study

| Original name           | New accepted name       | WFO ID         | Authorship |
|-------------------------|-------------------------|----------------|------------|
| Dipterocarpus conformis | Dipterocarpus conformis | wfo-0000651311 | Slooten    |
| Dryobalanops lanceolata | Dryobalanops lanceolata | wfo-0000657521 | Burck      |

| Original name         | New accepted name        | WFO ID         | Authorship                             |
|-----------------------|--------------------------|----------------|----------------------------------------|
| Hopea sangal          | Hopea sangal             | wfo-0000724645 | Korth.                                 |
| Parashorea malaanonan | Parashorea malaanonan    | wfo-0000396696 | (Blanco) Merr.                         |
| Parashorea tomentella | Parashorea tomentella    | wfo-0000396691 | (Symington) Meijer                     |
| Shorea argentifolia   | Rubroshorea argentifolia | wfo-1000050969 | (Symington) P.S.Ashton & J.Heck.       |
| Shorea beccariana     | Rubroshorea beccariana   | wfo-1000050992 | (Burck) P.S.Ashton & J.Heck.           |
| Shorea faguetiana     | Richetia faguetiana      | wfo-1000051017 | (F.Heim) P.S.Ashton & J.Heck.          |
| Shorea gibbosa        | Richetia gibbosa         | wfo-1000051019 | (Brandis) P.S.Ashton & J.Heck.         |
| Shorea johorensis     | Rubroshorea johorensis   | wfo-1000050945 | (Foxw.) P.S.Ashton & J.Heck.           |
| Shorea leprosula      | Rubroshorea leprosula    | wfo-1000050975 | (Miq.) P.S.Ashton & J.Heck.            |
| Shorea macrophylla    | Rubroshorea macrophylla  | wfo-1000050993 | (de Vriese) P.S.Ashton & J.Heck.       |
| Shorea macroptera     | Rubroshorea macroptera   | wfo-1000050964 | (Dyer) P.S.Ashton & J.Heck.            |
| Shorea ovalis         | Rubroshorea ovalis       | wfo-1000051055 | (Korth.) P.S.Ashton & J.Heck.          |
| Shorea parvifolia     | Rubroshorea ovata        | wfo-1000050977 | (Dyer ex Brandis) P.S.Ashton & J.Heck. |

### 3 Survival analysis of logged forest seedlings in their first census

57% of all individuals were never recorded as alive (5,868 out of 10,272) i.e., they died in the period between planting and their first census. This occurs exclusively in the logged forest since unlike in the old-growth forest, diameter measurements were not taken at the time of planting.

We excluded these individuals from the main analysis since they lack data both on size and exact age, hence the seedlings cannot be directly compared. For the first cohort of seedlings planted into the logged forest, between 18 and 20 months elapsed between planting and censusing, in that time 70% of seedlings died (5,034 out of 7,222). For the second cohort (replacing dead seedlings from cohort one), between 7 and 28 months elapsed between planting and censusing, and 30% died (834 out of 2,746).

Nevertheless, this data might provide some insight on which species are more likely to survive over these time periods, and we present a simple analysis here.

Survival of seedlings from cohort one and two were assessed in separate models with identical formulae. We fit the models in the R package `brms` with a Bernoulli response distribution, where zero indicated the seedling was dead in its first census and one indicated it was alive. The predictor was a fixed effect of species. We use a regularising prior of `normal(0, 2)`, a common choice for Bernoulli models. Results are presented in Figure 3 and Figure 4.

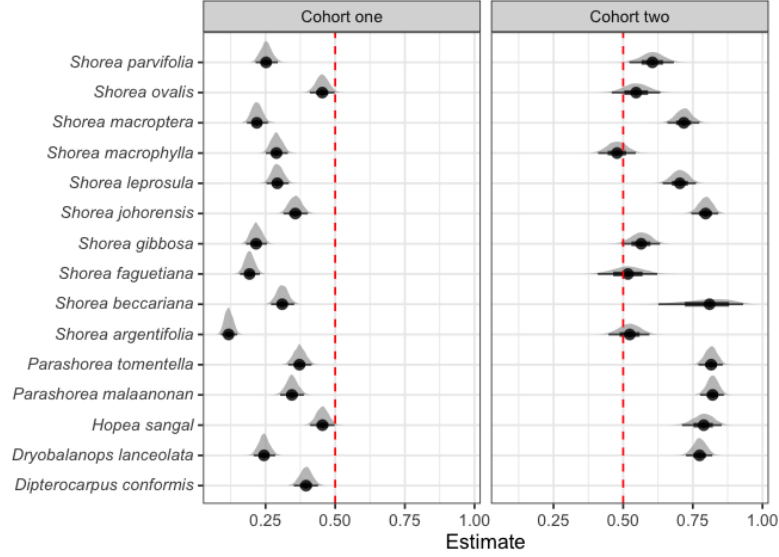


Figure 3: Posterior estimates of survival probability in the time period between planting and the first census of cohort one and two seedlings in the logged forest. The posterior median is represented by a point with the line showing the 66% and 95% credible interval. The dashed red line shows where the likelihood of survival is 50%. No *Dipterocarpus conformis* seedlings were planted in the second cohort due to insufficient availability.

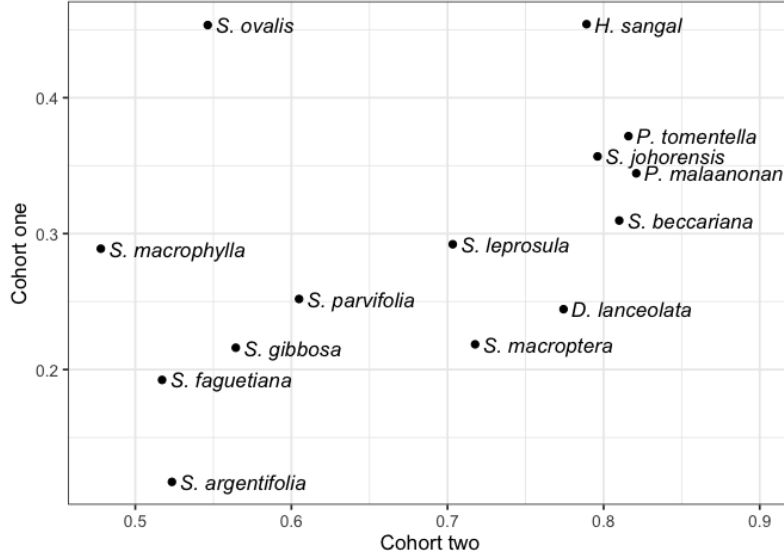


Figure 4: Median posterior estimates of survival probability for seedlings in cohort one ( $y$  axis) and two ( $x$  axis) in the time period between planting and their first census.

## 4 Estimating missing values of basal diameter

There are 190 missing values of basal diameter in our data (0.73% of all records for living trees). For these missing values, we estimated basal diameter from diameter at breast height using a known allometric equation,<sup>2</sup>

$$d_{h2} = \frac{D_{h1}}{\exp(b_1 h_1 - h_2)}$$

where  $d_{h2}$  is the estimated diameter (mm) at height  $h_2$  (m),  $D_{h1}$  is the known diameter (mm) at height  $h_1$  (m), and  $b_1$  is a taper parameter.

We chose the value for our taper parameter  $b_1 = 3.91$  by optimising on trees where we had diameter measurements both at the base and at breast height, to find the value which gave the lowest Root Mean Squared Error.

<sup>2</sup>Cushman, K. C. et al. Improving estimates of biomass change in buttressed trees using tree taper models. *Methods in Ecology and Evolution* 5, 573–582 (2014). doi: [10.1111/2041-210X.12187](https://doi.org/10.1111/2041-210X.12187)

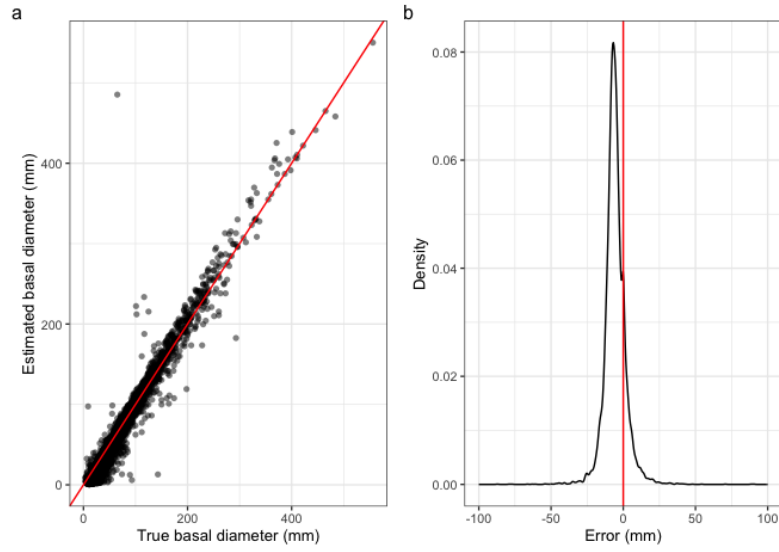


Figure 5: In (a), estimated values of basal diameter are plotted against their true values. The red line indicates where the estimate is equal to the true value. (b) Shows the distribution of the error around zero (red line).

## 5 Prior choice

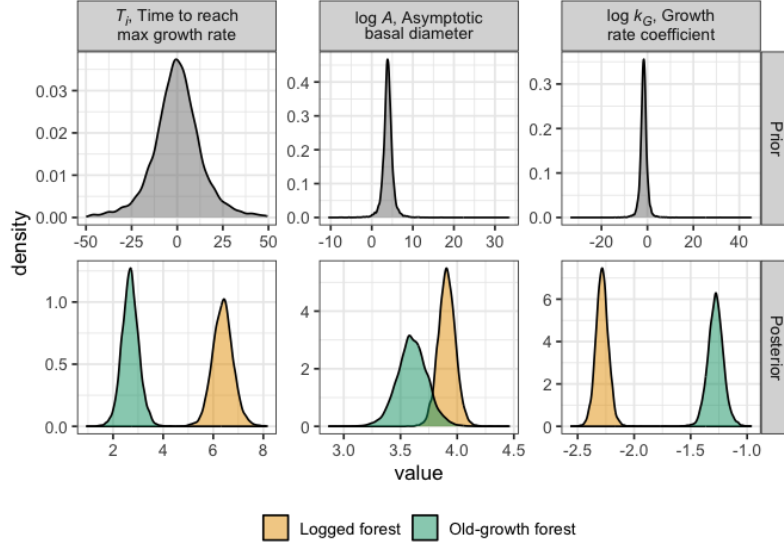


Figure 6: Prior and posterior distributions for the growth model on the link scale. Facet columns indicate parameters estimated by the model. Facet rows indicate the prior and posterior distributions. Note that the same priors were set for both forest types, and that axes are not preserved across panels.

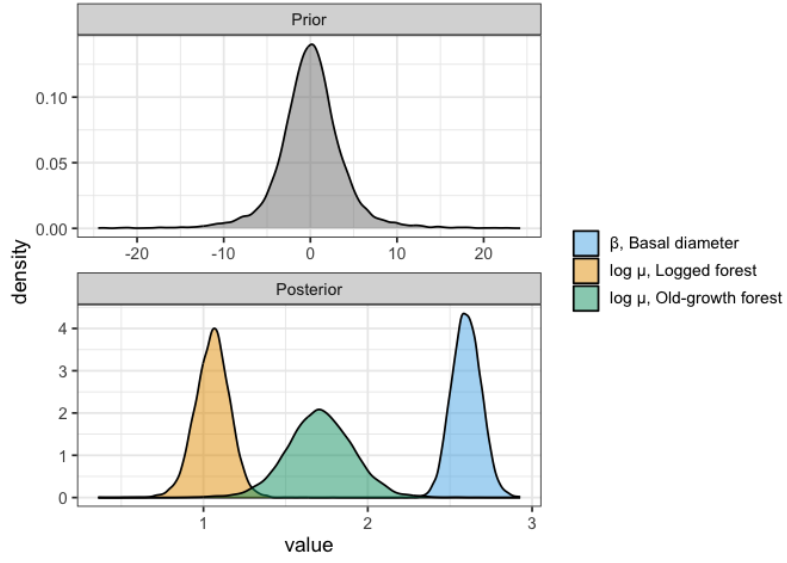


Figure 7: Prior and posterior distributions for the survival model on the link scale. Prior and posterior distributions are displayed in separate panels and colours indicate parameters estimated by the model. Note that the same prior was set across all the population-level parameters, and that axes are not preserved across panels.

## 6 Growth curves over time

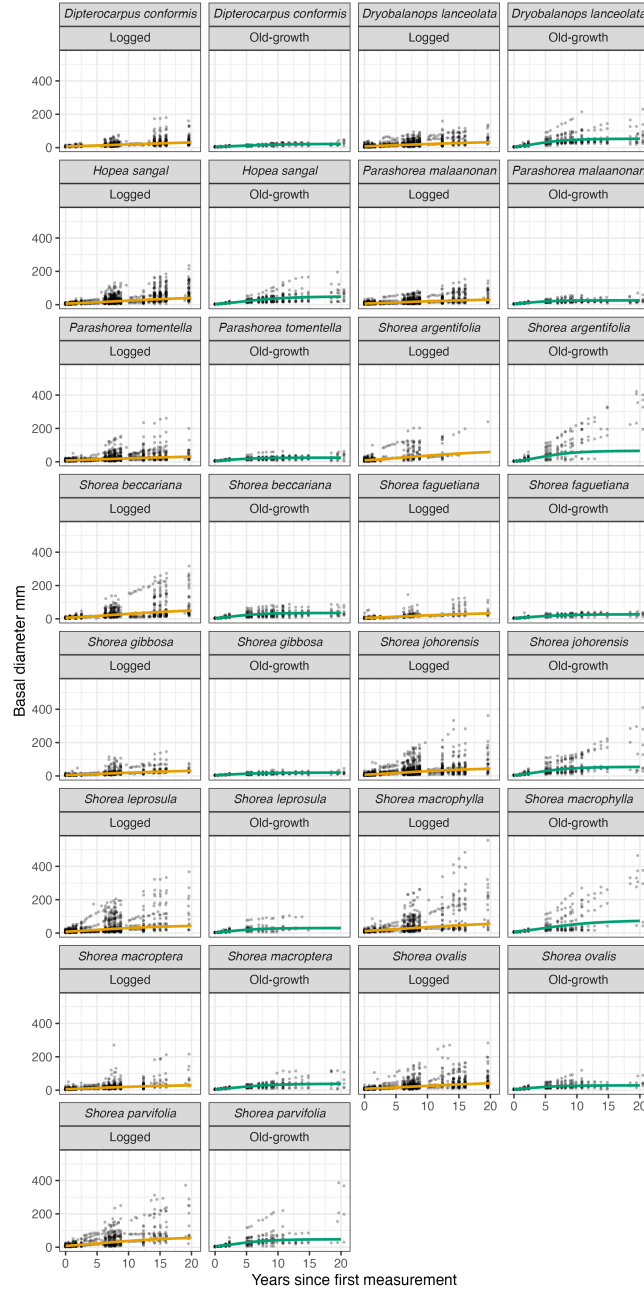


Figure 8: Basal diameter measurements for all seedlings over time (black points). Coloured lines show posterior predictions for a typical seedling of each of the 15 species in the old-growth (green) and logged (yellow) forest.

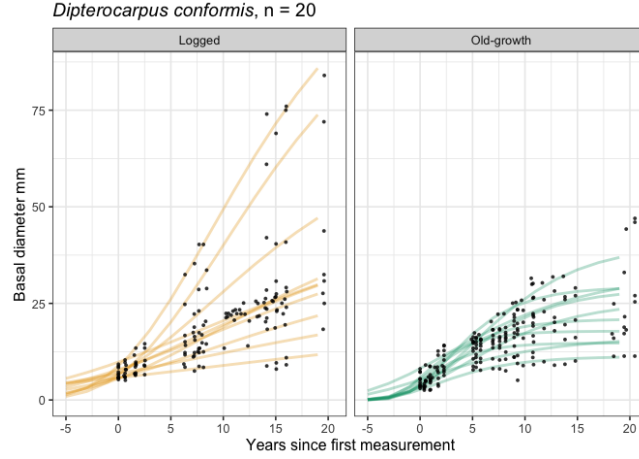


Figure 9: Basal diameter measurements for 20 randomly selected *Dipterocarpus conformis* seedlings over time (black points). Coloured lines show posterior predictions for individuals in the old-growth (green) and logged (yellow) forest.

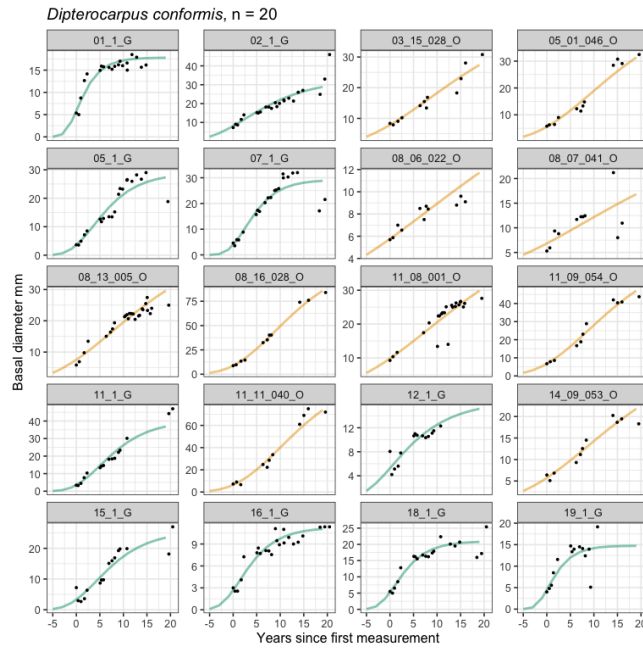


Figure 10: Basal diameter measurements for 20 randomly selected *Dipterocarpus conformis* seedlings over time (black points). These are the same individuals which are shown in Figure 9 above. Coloured lines show posterior predictions for individuals in the old-growth (green) and logged (yellow) forest. Note that the y-axis scale varies between panels.

## 7 Posterior predictive checks

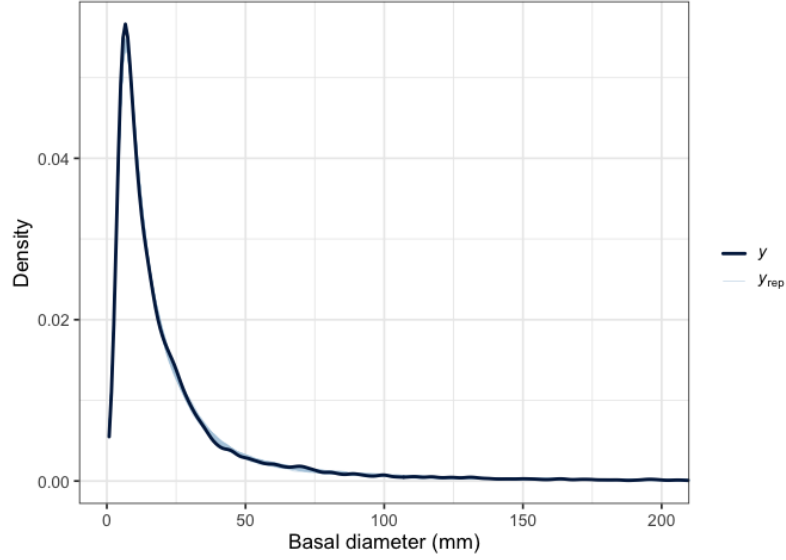


Figure 11: Posterior predictive check for the growth model, where the observed data is in dark blue ( $y$ ) and 50 draws from the posterior distribution are in light blue ( $y_{rep}$ ).

Table 2: Posterior estimates of population-level effects from the growth model on the link scale

| Pa-<br>rame-<br>ter  | Forest<br>type | Posterior<br>median | l-95%<br>CI | u-<br>95%<br>CI | Rhat  | Bulk effective<br>sample size | Tail effective<br>sample size |
|----------------------|----------------|---------------------|-------------|-----------------|-------|-------------------------------|-------------------------------|
| logA                 | Logged         | 3.909               | 3.757       | 4.061           | 1.001 | 5614.307                      | 6642.914                      |
| logA                 | Old-<br>growth | 3.604               | 3.336       | 3.876           | 1.000 | 7817.899                      | 6747.319                      |
| logk <sub>G</sub> /e | Logged         | -2.283              | -<br>2.394  | -<br>2.172      | 1.000 | 6115.659                      | 6917.982                      |
| logk <sub>G</sub> /e | Old-<br>growth | -1.279              | -<br>1.416  | -<br>1.137      | 1.000 | 9874.542                      | 6836.225                      |
| T <sub>i</sub>       | Logged         | 6.380               | 5.576       | 7.229           | 1.000 | 6780.500                      | 6621.473                      |
| T <sub>i</sub>       | Old-<br>growth | 2.689               | 2.021       | 3.397           | 1.000 | 9540.742                      | 7616.194                      |

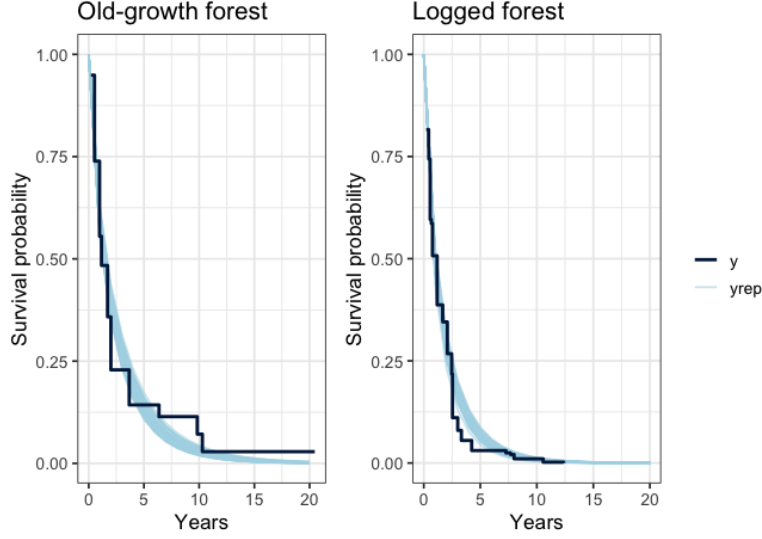


Figure 12: Posterior predictive check for the survival model, where the observed data is in dark blue ( $y$ ) and 50 draws from the posterior distribution are in light blue ( $yrep$ ). Conditional Kaplan-Meier curves, represented by  $y$ , were estimated for seedlings at representative values of basal diameter (the 0.25, 0.5 and 0.75 quartiles  $\pm$  the standard distribution multiplied by 0.01). Posterior predictions ( $yrep$ ), were made for seedlings of these exact sizes (the 0.25, 0.5 and 0.75 quartiles).

Table 3: Posterior estimates of population-level effects from the survival model on the link scale

| Parameter                         | Posterior<br>median | l-95%<br>CI | u-<br>95%<br>CI | Rhat  | Bulk effective<br>sample size | Tail effective<br>sample size |
|-----------------------------------|---------------------|-------------|-----------------|-------|-------------------------------|-------------------------------|
| $\log \mu$ , Logged<br>forest     | 1.057               | 0.842       | 1.255           | 1.001 | 2090.704                      | 3589.735                      |
| $\log \mu$ , Old-growth<br>forest | 1.708               | 1.307       | 2.106           | 1.000 | 2583.691                      | 4432.720                      |
| $\beta$ , Basal diameter          | 2.599               | 2.432       | 2.775           | 1.000 | 10644.645                     | 7872.663                      |

## 8 Growth-survival correlations

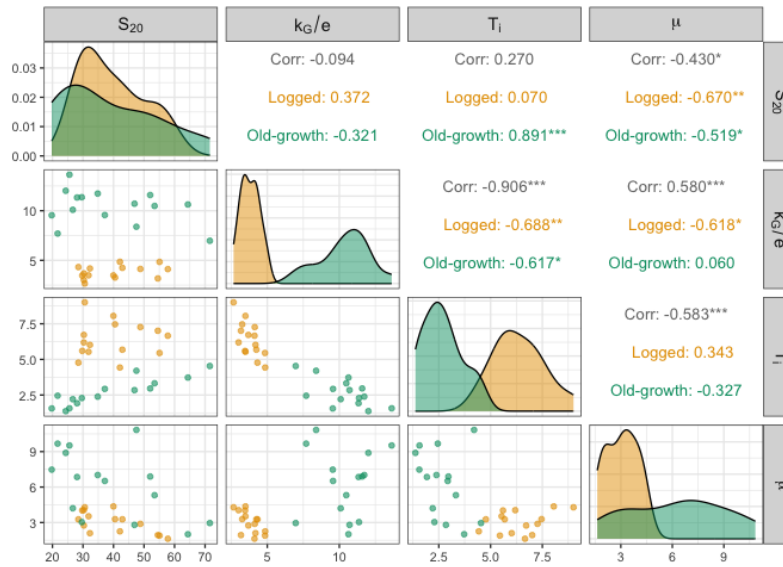


Figure 13: Pairwise Pearson's correlation coefficients for growth and survival parameters.